Finite-memory projection corrections as a diagnostic gate for cosmological consistency tests

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Abstract

We introduce a bounded finite-memory projection operator for diagnostic cosmological consistency tests. The operator is formulated as an effective projection response over normalized depth, $W(x) = 1 + \rho x^3$, with a passive memory-budget bound $\rho \leq 4$. The construction is not presented as a complete cosmological model or a direct measurement claim. It is instead a disciplined diagnostic module for comparing projected response shapes against backreaction-aware distance consistency reconstructions. In a BAO-only diagnostic reconstruction, the locked K2 window overlaps the full target envelope. In an SN+BAO source-split stress test, strict median-sign matching exhibits a localized tension; a reconstruction-aware sign-stability gate keeps this tension classified as non-violating without widening the K2 window. The result motivates a future full-likelihood comparison while preserving explicit claim boundaries.

1. Introduction

Modern cosmological consistency tests increasingly distinguish background expansion, light-propagation effects, curvature consistency, and cosmic backreaction. A residual relative to a homogeneous FLRW reference model is therefore not automatically evidence for new physics: it may arise from optical sampling, reconstruction convention, averaging, or the choice of observable combination.

Heinesen and Clifton recently emphasized that broad classes of cosmological models can be separated by how they violate FLRW curvature-consistency tests [1]. They isolate two especially relevant routes: deviations in optical properties and deviations in large-scale expansion. This is naturally adjacent to the older Dyer-Roeder optical-distance tradition, where light propagation through clumpy matter distributions is treated separately from the homogeneous background distance law [3,4]. Koksbang then applied this kind of observable logic to constrain cosmic backreaction over an extended redshift range, finding consistency with vanishing backreaction at one standard deviation while also emphasizing that significant backreaction is not yet ruled out [2]. These papers are useful here because they make the main methodological lesson sharp: a projected cosmological residual must be tested against optical, geometrical, and reconstruction degeneracies before it is interpreted.

This note develops a deliberately narrow finite-memory projection diagnostic that can be placed alongside such tests. The objective is not to validate a full physical model. Instead, we ask whether a predeclared, bounded projection-memory operator can define a reproducible diagnostic window and survive basic source-split stress tests.

The contribution is therefore threefold. First, we define a finite-memory operator whose amplitude is bounded before comparison with reconstructed diagnostics. Second, we give an explicit shape-selection rule that fixes the cubic kernel from a small power-kernel family. Third, we state a sign-stability policy for interpreting reconstructed diagnostic points whose median sign is method-sensitive.

2. Relation To Existing Diagnostic Ideas

The present construction is closest in spirit to a consistency diagnostic, not to a replacement background cosmology. In a standard FLRW interpretation, distances, expansion rates, and curvature relations are locked together by the assumed homogeneous geometry. A violation of one of these relations can point to several different mechanisms: altered large-scale expansion, altered light propagation, averaging effects, or reconstruction artifacts. This is why the same residual pattern can be read very differently depending on whether one is testing optical propagation, backreaction, or modified dynamics.

The finite-memory operator introduced here is designed to sit at this interface. It does not start by changing the background Friedmann equations. Instead, it asks whether a projected diagnostic response can carry a bounded memory of accumulated depth. This makes it naturally comparable to Dyer-Roeder-type optical reasoning, where light propagation through clumpy matter is separated from the homogeneous reference law, and to modern backreaction tests, where averaged expansion and observed distance relations need not encode the same effective geometry.

The important difference is that the operator below is not an additional free-form fitting function. Its amplitude and shape are constrained before the diagnostic comparison. This is the main methodological point: the operator is allowed to be phenomenological, but it is not allowed to become arbitrarily elastic after seeing the reconstructed envelope.

For this reason, the most important object in the paper is not the specific cubic formula by itself. The stronger contribution is the combination of a bounded response window, a predeclared admissibility rule, and a reconstruction-stability-aware interpretation policy.

3. Effective Diagnostic Module

Only the effective module needed for this note is used. We assume that an observed consistency residual can be represented as a projected response over a normalized depth coordinate,

$$0 \leq x \leq 1.$$

The variable x should be read operationally: it orders the accumulated projected response along the diagnostic light-path/reconstruction depth. In the point-level diagnostic table below, the public packet uses the frozen operational mapping

$$x = z/z_{\max}$$

over the tested SN+BAO redshift grid. This is a redshift-ordered diagnostic coordinate, not a final physical identification of x with redshift and not a new cosmological observable.

The finite-memory correction is a multiplicative response operator acting on a baseline projected shape:

$$K_2(x) = W(x)K_1(x).$$

This note studies the locked K2 form

$$W(x) = 1 + \rho x^3.$$

The form is intentionally simple. Its purpose is to test whether a bounded, delayed, monotone memory response can be made reproducible before comparison with diagnostic envelopes.

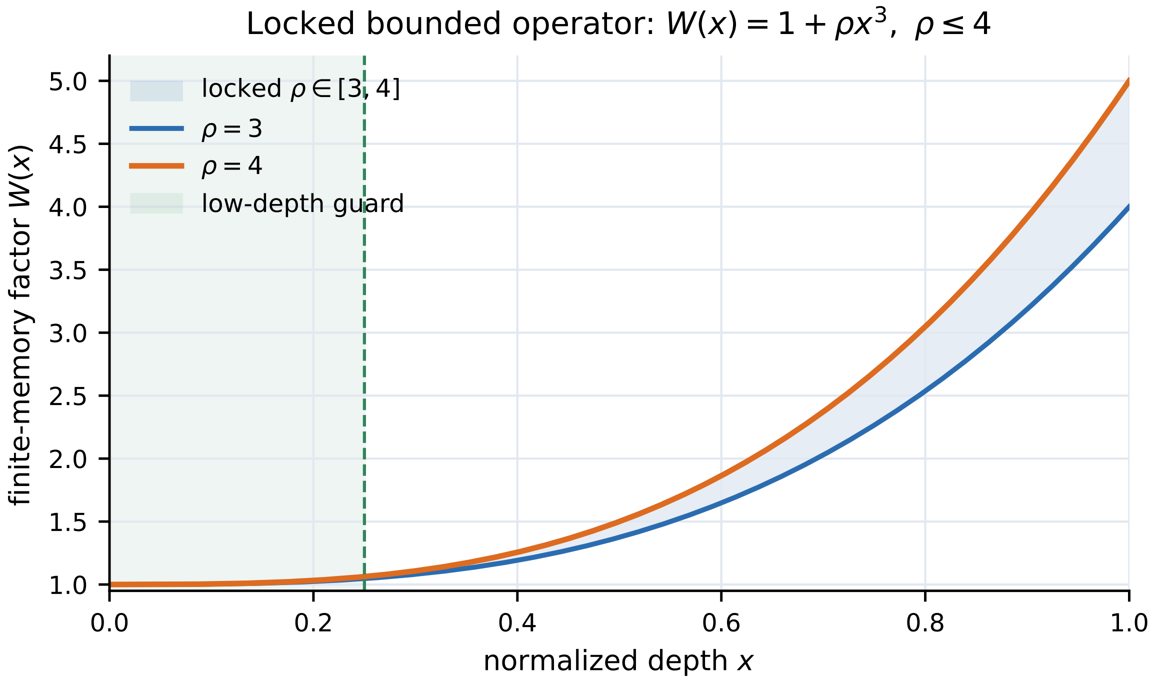

Figure 1: Locked finite-memory operator. The shaded band is the predeclared $\rho \in [3, 4]$ window inside the passive bound $\rho \leq 4$; the low-depth guard shows why the response remains weak near the origin.

Robustness And Coordinate Dependence

The present note is coordinate-explicit. Its numerical tables use

$$x = z/z_{\max}$$

as the current frozen operational diagnostic mapping. This choice is useful because it is simple, reproducible, and tied directly to the public redshift grid. It is not claimed to be a coordinate-invariant or physically final description of cosmological depth.

Two natural robustness mappings are left for the next stage:

$$x = \chi(z)/\chi(z_{\max}),$$

where χ is a comoving-distance coordinate under a frozen reference cosmology, and an optical-depth-like coordinate in which the depth variable is weighted by foreground or visibility information. A likelihood-native mapping, using the reconstruction grid or likelihood coordinate directly, is the Phase II target. Until such mappings are implemented, the method note should be read as a coordinate-explicit diagnostic proposal only.

K1 Baseline Provenance

The baseline shape $K_1(x)$ is a frozen diagnostic baseline imported from the distilled reconstruction packet. It is not fitted in this note. The comparison is organized around whether the locked multiplicative memory factor can keep the projected K2 response inside the reconstructed diagnostic envelope under the predeclared amplitude and shape rules.

The present manuscript therefore tests the finite-memory multiplier on top of a fixed diagnostic baseline. It does not establish that $K_1(x)$ is unique, physically final, or likelihood-preferred.

Several assumptions are therefore intentionally modest:

- the coordinate x is a diagnostic ordering variable, not a new directly measured cosmological coordinate;
- the operator $W(x)$ is an effective response, not a complete field equation;
- the reconstructed envelopes are treated as diagnostic targets, not as final likelihood surfaces;
- the public module does not require the reader to accept any broader background theory.

These restrictions make the paper weaker than a full theory paper, but stronger as a falsifiable method note. A reader can reject or test the effective module without needing to adjudicate any larger research program.

4. Passive Memory Bound

The passive memory-budget condition requires the accumulated tail response not to exceed the unit local response budget. For the cubic operator,

$$E_{\rm tail} = \int_0^1 \rho x^3 \, dx = \frac{\rho}{4}.$$

The unit baseline budget is

$$E_0 = \int_0^1 1 \, dx = 1.$$

Passivity requires

$$E_{\rm tail} \le E_0, \qquad \frac{\rho}{4} \le 1, \qquad \rho \le 4.$$

This is the key discipline of the construction: the memory depth is bounded before the diagnostic comparison. The bound is not chosen by looking for the best residual fit.

The word "passive" is used in this limited sense only. It means that the memory tail does not carry more integrated response budget than the local baseline term over the normalized interval. It does not mean that the underlying cosmology is static, nor that no physical energy exchange is possible in a deeper model. For this paper, passivity is simply a guard against an uncontrolled late-depth amplification.

This bound also gives an immediate falsification handle. If a diagnostic comparison requires $\rho > 4$ in order to remain compatible with the target envelope, the locked passive operator is not supported in its present form. The model is not then rescued by expanding the allowed amplitude range.

5. Kernel-Shape Selection

The cubic kernel is selected from the power-kernel family

$$W_p(x) = 1 + \rho x^p.$$

For this family, the passive memory rule generalizes to

$$\int_0^1 \rho x^p \, dx \le \int_0^1 1 \, dx, \qquad \frac{\rho}{p+1} \le 1, \qquad \rho \le p + 1.$$

Two additional effective-operator rules select the usable shape:

$$\Delta W_p(0.25) = (p+1)0.25^p \le 0.10,$$

$$\int_{0.75}^1 (p+1)x^p \, dx \le 0.75.$$

The first rule rejects kernels that disturb the already locked low-depth response too early. The second rejects kernels whose memory budget collapses too sharply onto the last quarter of the path. In the present audit, p=1 and p=2 are too visible at low depth, while p>=4 is too endpoint-concentrated. The cubic p=3 case is the remaining admissible power-kernel representative.

The locked diagnostic window is therefore

$$W(x) = 1 + \rho x^3, \qquad \eta \in [0.25, 0.50], \qquad \rho \in [3.00, 4.00].$$

The two shape rules encode a compromise. A linear or quadratic memory term becomes visible too early and risks contaminating the low-depth regime where the baseline diagnostic response should remain dominant. Higher powers delay the response more strongly, but at the cost of concentrating the correction too close to the endpoint. The cubic term is therefore not claimed to be uniquely fundamental; it is the lowest simple power that is late enough at low depth while not being too endpoint-like.

This distinction matters for interpretation. The paper does not say that cosmology ”must” use a cubic memory law. It says that, inside this deliberately small power-kernel audit, the cubic law is the locked representative that survives the predeclared admissibility rules.

The numerical thresholds in these two rules should also be read conservatively. They are method thresholds for the present diagnostic packet, not universal constants. A later likelihood paper should report threshold-sensitivity runs showing whether the cubic representative remains selected when the low-depth visibility and endpoint-concentration cuts are varied within a reasonable predeclared range.

The current threshold-sensitivity policy is summarized as follows:

Threshold set	Low-depth limit	Endpoint budget	Admissible simple kernels	Interpretation
baseline	$\Delta W_p(0.25) \le 0.10$	endpoint ≤ 0.75	$p = 3$	cubic is the current audit-selected representative
relaxed low-depth	$\Delta W_p(0.25) \le 0.15$	endpoint ≤ 0.75	$p = 2, 3$	more low-depth visibility broadens the set
strict low-depth	$\Delta W_p(0.25) \le 0.075$	endpoint ≤ 0.75	$p = 3$	cubic remains in this simple scan
relaxed endpoint	$\Delta W_p(0.25) \le 0.10$	endpoint ≤ 0.85	$p = 3, 4$	endpoint relaxation admits a later kernel
strict endpoint	$\Delta W_p(0.25) \le 0.10$	endpoint ≤ 0.60	none	the simple power-kernel family can be emptied

This means the cubic kernel is not a fundamental cosmological law. It is the baseline representative selected by the current audit thresholds. Relaxing or tightening those thresholds can broaden, shift, or empty the admissible simple power-kernel set.

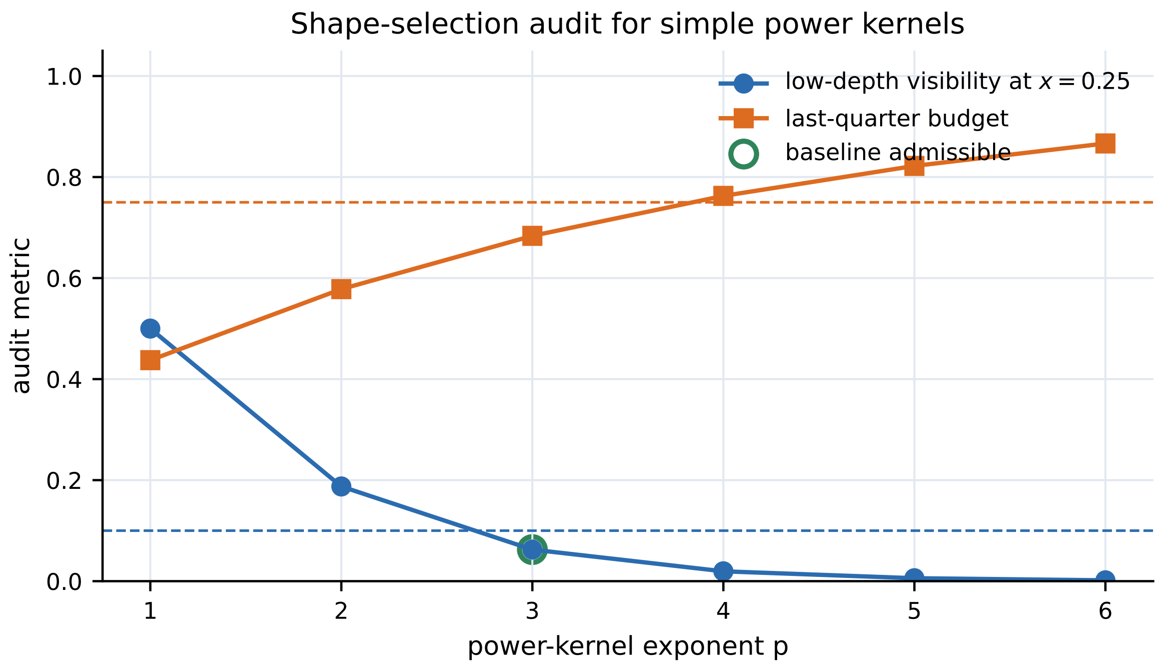

Figure 2: Power-kernel shape-selection audit. The cubic representative is selected by the current low-depth visibility and endpoint-budget thresholds; this is a methodological selection rule, not a fundamental-law claim.

6. Diagnostic Targets And Gate Policy

The targets used here are reconstructed diagnostic envelopes rather than direct published likelihoods. The comparison is therefore deliberately separated into three levels:

```
sign-stable reconstructed point   -> exact sign gate
sign-unstable reconstructed point -> envelope / diagonal residual gate
controls and extrapolations       -> stress diagnostics only
```

Operationally, let y_i denote the reconstructed diagnostic median at the tested point i, and let $[L_i, U_i]$ denote the corresponding reconstructed diagnostic envelope. A locked prediction $\hat{y}_i$ is envelope-compatible when

$$L_i \leq \hat{y}_i \leq U_i.$$

If a diagonal diagnostic width σ_i is available, the normalized residual gate is

$$|\hat{y}_i - y_i| / \sigma_i \leq 1.$$

A reconstructed point is sign-stable only when the relevant method/degree families agree on the sign of the diagnostic median. If the reconstruction sign is method-sensitive, the median sign is not used as a hard falsifier. Instead, the locked K2 prediction must remain inside the diagnostic envelope and have a normalized residual not exceeding one standard diagnostic width.

This policy is not a substitute for a full likelihood. It is a triage rule that prevents a method-sensitive reconstructed median from being overinterpreted.

This distinction is important for the SN+BAO stress test below. A strict median-sign gate asks whether the prediction follows the sign of one reconstructed median. The sign-stability gate asks a weaker but better-posed question: whether the prediction violates a sign that is stable across reconstruction choices. Only the second question is used as a paper-level diagnostic violation/non-violation rule here.

7. Source-Split Interpretation

The BAO-only and SN+BAO diagnostics are not interchangeable. BAO-only reconstructions are comparatively closer to a distance-scale consistency test, while SN+BAO combinations mix distance ladder information and reconstruction choices in a way that can sharpen or destabilize local features. A finite memory operator should therefore not be judged only by whether it follows a single reconstructed median. It should also be judged by whether a claimed tension is robust across reconstruction families.

In the present packet, this is exactly where the localized SN+BAO warning appears. The strict median-sign comparison identifies a point near z=1.1925 where the median sign is negative while the locked K2 response does not follow that sign. The reconstruction audit, however, indicates that this sign is not stable across the tested method/degree families. Lower-degree reconstructions favor the opposite sign, while higher-degree reconstructions pull the median negative.

The sign-stability policy is intended to prevent this point from being used in either direction too aggressively. It should not be hidden, because it is a real stress point for the diagnostic. It should also not be promoted to a hard falsification, because the sign being tested is reconstruction-sensitive. The paper-level statement is therefore deliberately asymmetric: the point remains a warning, but not a locked diagnostic violation under the envelope and normalized-residual gate.

8. Results

The minimal result summary is:

Diagnostic	Result	Interpretation
BAO-only window overlap	1.0000000000	Locked K2 overlaps the BAO-only diagnostic envelope at all tested points.
BAO-only best curve overlap	1.0000000000	The best locked K2 BAO-only curve remains inside the envelope.
SN+BAO sign-stable non-violation fraction	1.0000000000	All sign-stable SN+BAO points are non-violating under exact sign matching.
SN+BAO sign-unstable compatibility fraction	1.0000000000	All sign-unstable SN+BAO points remain envelope-compatible under diagonal residual handling.

The strict SN+BAO median-sign test exhibits a localized tension near z=1.1925. The diagnostic audit shows that this point is method-degree sensitive: lower-degree reconstructions are positive, while higher-degree reconstructions pull the median negative. The locked K2 prediction does not follow the negative median sign at that point, but it remains inside the broad diagnostic envelope with a small normalized residual. Under the sign-stability gate, this is treated as an envelope-compatible warning rather than a hard sign violation.

The current status is therefore:

```
BAO-only: diagnostic compatibility
SN+BAO strict median-sign: localized warning
SN+BAO sign-stability-aware gate: non-violating diagnostic status
claim level: theory-method diagnostic, not measurement validation
```

The point-level SN+BAO audit makes the gate policy more explicit. The source for all rows is the no-degree-4 robust SN+BAO envelope, and the x column uses the provisional mapping $x = z/z_{\max}$ over this grid.

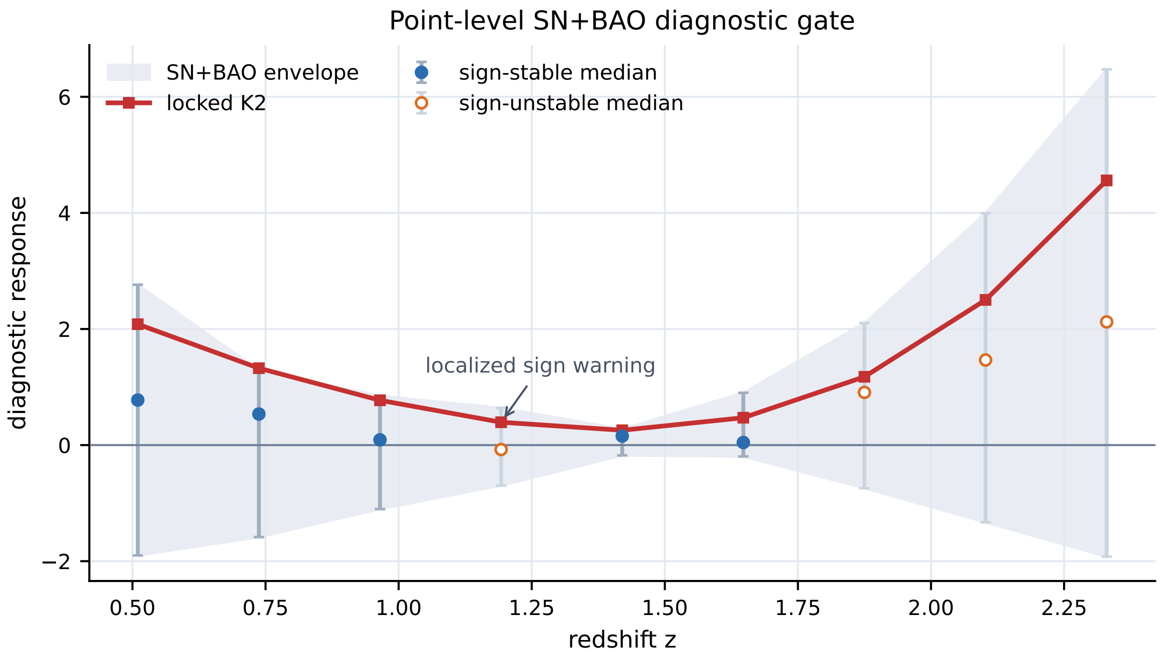

Figure 3: Point-level SN+BAO diagnostic gate. Filled points are sign-stable reconstructed medians, open points are sign-unstable medians, the grey band is the no-degree-4 robust envelope, and the red curve is the locked K2 prediction.

z	x	median	envelope	K2	signs all/no4/d2	stable?	residual/sigma	decision
0.5100	0.2189	0.7758	[-1.9013, 2.7631]	2.0829	+/+/+	yes	-0.0350	NONVIOLATING_SIGN_STABLE
0.7375	0.3165	0.5350	[-1.5846, 1.2890]	1.3251	+/+/+	yes	-0.0565	NONVIOLATING_SIGN_STABLE
0.9650	0.4142	0.0906	[-1.1023, 0.8498]	0.7717	+/+/+	yes	0.1417	NONVIOLATING_SIGN_STABLE
1.1925	0.5118	-0.0762	[-0.6967, 0.6401]	0.3938	-/-/+	no	0.1971	COMPATIBLE_SIGN_UNSTABLE
1.4200	0.6094	0.1560	[-0.1773, 0.2784]	0.2554	+/+/+	yes	0.0421	NONVIOLATING_SIGN_STABLE
1.6475	0.7071	0.0445	[-0.1959, 0.9010]	0.4717	+/+/+	yes	0.2289	NONVIOLATING_SIGN_STABLE
1.8750	0.8047	0.9081	[-0.7426, 2.1026]	1.1769	+/+/-	no	-0.1356	COMPATIBLE_SIGN_UNSTABLE
2.1025	0.9024	1.4662	[-1.3295, 3.9935]	2.5013	+/+/-	no	-0.0636	COMPATIBLE_SIGN_UNSTABLE
2.3300	1.0000	2.1225	[-1.9217, 6.4719]	4.5595	+/+/-	no	0.0079	COMPATIBLE_SIGN_UNSTABLE

This table also makes the main weakness visible. The non-violation decisions are not equivalent to a strong likelihood preference: several envelopes are broad, and four of the nine SN+BAO points are sign-unstable. The table supports the narrow claim that the locked K2 response does not violate the current diagnostic gate; it does not support a direct measurement claim.

9. Measurement Gate Preflight Status

The measurement-gate scaffold is operational, but it remains a preflight benchmark rather than a measurement-validation result. The purpose of the gate is to keep the finite-memory diagnostic ansatz testable: the K2 operator is locked before scoring, null comparators are registered in advance, and stronger claims remain blocked until covariance-aware public benchmarks and calibrated physical nulls are available.

9.1 Preflight Status

The current implementation shows five relevant facts. First, the BAO-only diagnostic envelope remains compatible with the locked K2 window under the current reconstruction-level packet. Second, the SN+BAO comparison contains a localized sign-stability warning rather than a clean median-sign match. Third, coordinate robustness is not automatic: an operator-only chi-normalized mapping produces an envelope-tension warning, although bounded rho values within the allowed rho <= 4 range can recover non-violating status in that diagnostic stress test. Fourth, K1/no-memory remains competitive, and in some diagonal or public-proxy scorecards it is stronger than fixed K2. Fifth, several source-split and predeclared-protocol-guided preflight routes favor locked K2 over local reproduction-family and backreaction-proxy controls, but these are not likelihood-native measurement validations.

9.2 Coordinate Robustness

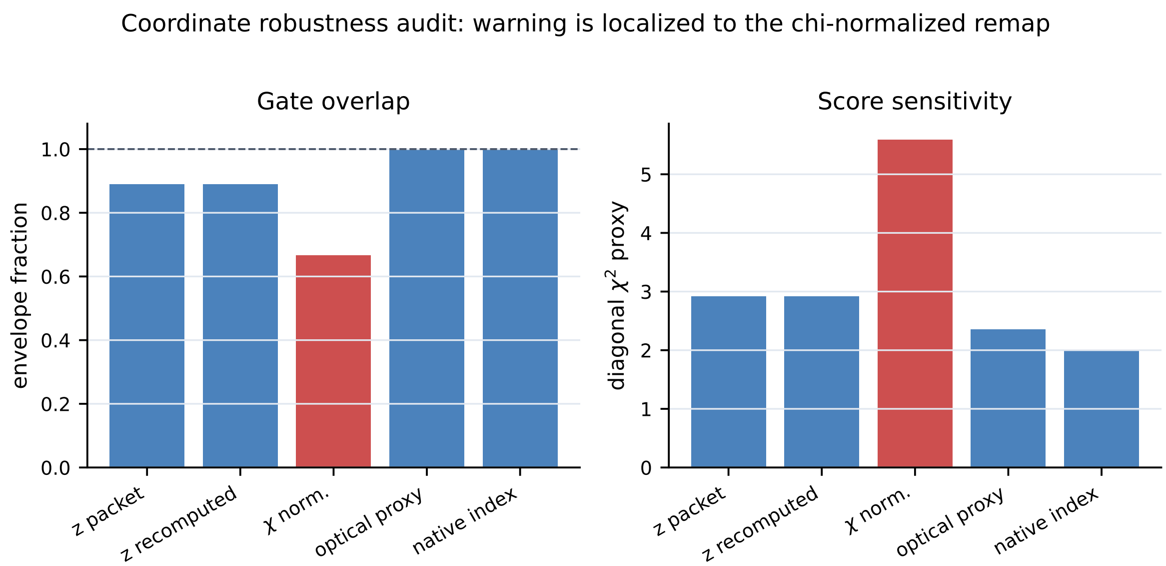

Figure 4: Coordinate robustness audit. Most mappings remain non-violating under the current gate, while the chi-normalized operator-only remap produces the localized coordinate warning that remains a Phase II target.

9.3 Null Comparators

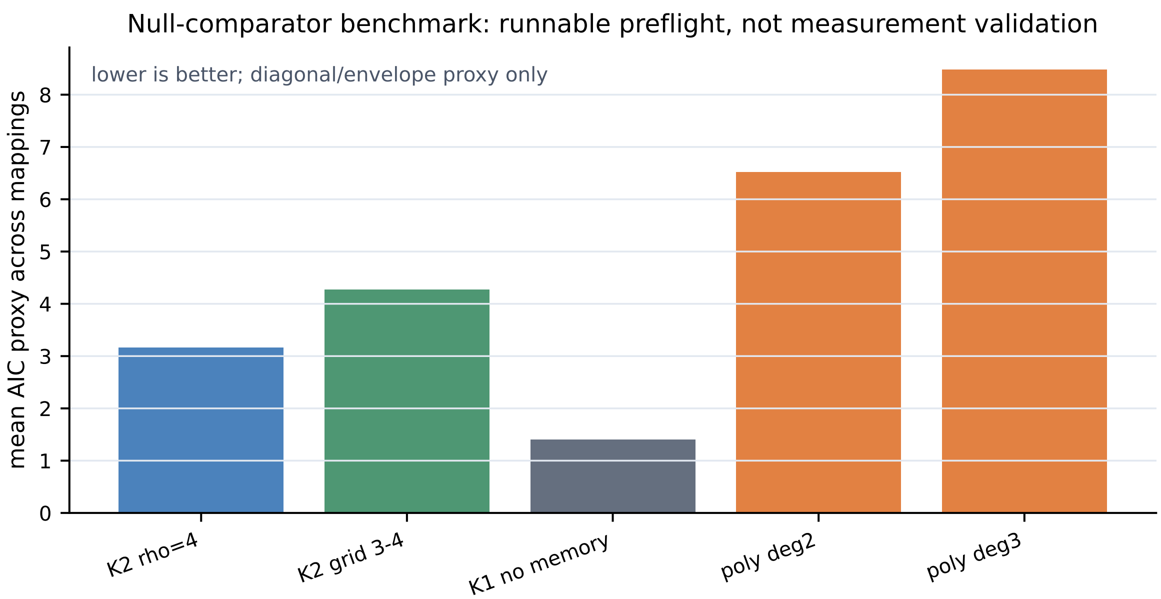

Figure 5: Null-comparator preflight scorecard. The chart reports proxy information-criterion scores across the current mappings; it is a benchmark diagnostic, not a measurement-validation result. These proxy scores are route-dependent and do not constitute likelihood-level model selection.

9.4 Gate Closure Conditions

Thus the measurement gate is useful precisely because it does not promote the current result too far. The present package supports a disciplined diagnostic claim: a bounded finite-memory operator can be locked, scored, compared with nulls, and stress-tested reproducibly. It does not yet support a measurement claim. The gate remains closed until the following objects are available:

- a likelihood-native K1/no-memory target;
- a joint SN+BAO covariance or a preregistered shrinkage-covariance benchmark;
- calibrated optical and backreaction physical-null comparators;
- source-native or independently reproduced derivative/covariance exports for the backreaction route;
- a frozen public benchmark in which no K2 kernel, rho bound, or K1 baseline is changed after scoring.

The detailed implementation log, including source-split export guards, covariance registries, predeclared-protocol-guided reproduction families, and backreaction preflight audits, is retained in the public repository as docs/measurement_gate_audit_log.md.

10. Robustness And Falsification Criteria

The current evidence is deliberately weaker than a full likelihood. For that reason, it is useful to state what would make the finite-memory diagnostic fail or become substantially less interesting.

The diagnostic would fail in its present form if:

- the BAO-only envelope overlap disappeared under the same locked K2 window;
- sign-stable SN+BAO points required the opposite sign from the locked prediction;
- compatibility required $\rho > 4$ or a different kernel power chosen after seeing the envelope;
- the localized SN+BAO warning became sign-stable under a stronger reconstruction audit;
- a full covariance likelihood rejected the whole locked window rather than merely shifting the best point inside it.

Conversely, the diagnostic would become more interesting if the same locked window survived a covariance-aware likelihood test, if independent reconstructions selected a similar late-depth response shape, or if backreaction-aware and optical-propagation tests separated in a way naturally tracked by the K2 window.

These criteria are included to keep the method honest. The finite-memory operator is useful only if it remains constrained. If it must be repeatedly relaxed to follow each new diagnostic reconstruction, it loses the main advantage of being predeclared.

11. Expected Significance

The main significance of the construction is not that it establishes a new cosmological model. Its value is methodological:

- it gives a compact finite-memory operator whose amplitude is bounded before comparison with data;
- it interfaces naturally with FLRW consistency, optical-propagation, and backreaction tests;
- it demonstrates a way to handle sign-unstable reconstructed diagnostics without silently overclaiming;
- it provides a small public module that can be tested independently as a diagnostic proposal.

The near-term audience is likely narrow: researchers interested in cosmological consistency tests, backreaction, light propagation, and method-level modified-gravity phenomenology. The higher-impact route requires a later full covariance or shrinkage-covariance likelihood comparison with direct public likelihood products.

12. Limitations

This manuscript is a theory-method diagnostic note. It does not claim:

- a full covariance likelihood;
- a direct measurement claim;
- exclusion of cosmic backreaction;
- a complete derivation of cosmology from an underlying action;
- reconstruction of a full background theory.

The next stage is a paper-aligned likelihood comparison using covariance or shrinkage-covariance products and direct published likelihood data where available.

The most important limitation is that the public packet contains distilled diagnostic summaries rather than the full reconstruction pipeline. This is acceptable for a method note whose purpose is to define a bounded operator and state claim boundaries, but it is not sufficient for an observational paper. The current manuscript should therefore be read as a bridge from a compact diagnostic construction to a public, testable effective module, while leaving the stronger likelihood comparison for a later release.

13. Phase II Roadmap

The next technical step is to replace the diagonal/envelope diagnostic with a likelihood-level comparison. At minimum, this requires public likelihood or data products, a covariance or shrinkage-covariance prescription, and a frozen mapping from the reconstructed observable to the finite-memory diagnostic coordinate x.

A useful future workflow is:

```
freeze K2 window -> ingest public likelihood products -> build covariance-aware
diagnostic -> compare locked window -> report posterior support or rejection
```

This future comparison should not widen the K2 window unless a new paper is explicitly framed as a different model. The main value of the present note is that it fixes a small target before the stronger test is performed.

The Phase II work package should therefore include:

- covariance-aware likelihood ingestion;
- null comparators for no-memory, smooth-response, optical-propagation, and backreaction-only alternatives;
- a public reconstruction benchmark;
- coordinate-mapping robustness across redshift-normalized, comoving-distance, optical-depth-like, and likelihood-native coordinates;
- direct comparison against BAO-only and SN+BAO reconstruction families;
- a frozen benchmark challenge in which the K2 window is locked before testing.

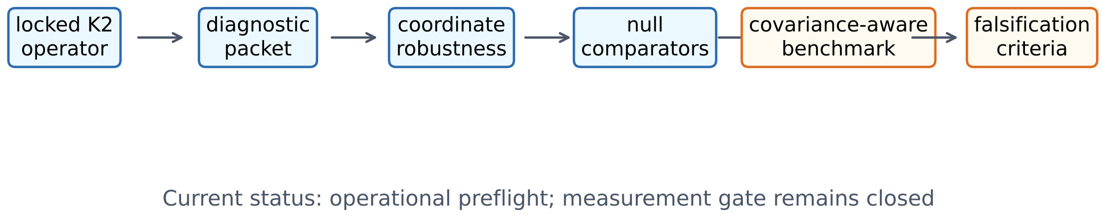

Figure 6: Measurement-gate preflight workflow. The current repository implements the left-hand preflight stages; the covariance-aware benchmark and falsification criteria remain the required route to stronger empirical claims.

14. Independent Wolfram Verification

The main diagnostic packet is generated by the Python gate engine, but the repository also includes an independent Wolfram Language audit: wolfram/FiniteMemory_Diagnostic_Check.wl. This is a second implementation check rather than an additional measurement analysis. It verifies the symbolic finite-memory identities

$$W(x) = 1 + \rho x^3, \qquad E_{\text{tail}} = \int_0^1 \rho x^3 \, dx = \frac{\rho}{4}, \qquad \rho \le 4,$$

and the generalized power-kernel relation

$$W_p(x) = 1 + \rho x^p, \qquad E_{\text{tail},p} = \frac{\rho}{p+1}, \qquad \rho \le p + 1.$$

The same Wolfram script then imports the compact manuscript-facing CSV outputs for the shape-selection audit, threshold sensitivity, and SN+BAO point-level gate. In the current repository state it reports ReproducibilityStatus=REPRODUCIBILITY_OK: the baseline shape-selection row selects p=3, the threshold-sensitivity table has the expected sensitivity rows, and the SN+BAO point-level gate contains nine rows with five sign-stable and four sign-unstable points under the declared decision policy.

The Wolfram audit is intentionally narrow. It does not replace the Python pipeline, does not ingest source-native likelihood products, and does not upgrade the claim level. Its role is reproducibility hygiene: the memory-budget formulae and the paper-facing diagnostic tables can be checked by a second implementation using a different computational environment. The audit checks reproducibility of the published formulae and tables, not the physical correctness of the finite-memory diagnostic ansatz.

15. Conclusion

We have defined a finite-memory projection correction as a bounded diagnostic operator rather than as a complete cosmological model. The locked cubic form $W(x) = 1 + \rho x^3$ follows from a passive memory budget and a simple shape-selection audit over power kernels. With $\rho \leq 4$, the operator gives a compact K2 diagnostic window that can be compared against reconstructed cosmological consistency envelopes without fitting the memory depth after the fact.

Within the distilled diagnostic packet used here, the locked K2 window is compatible with the BAO-only envelope and remains non-violating under the reconstruction-aware SN+BAO sign-stability gate. The strict SN+BAO median-sign warning is retained as a limitation rather than hidden: it marks a reconstruction-sensitive point that should be revisited with direct likelihood products.

The result is therefore a release-candidate method-note claim, not an measurement-validation claim. Its next meaningful test is a covariance-aware likelihood comparison against public likelihood data and predeclared null comparators.

16. Data And Code Availability

The public Paper 5 repository is hosted at https://github.com/tau-core-research/finite-memory-cosmology-paper5. It contains the manuscript source, rendered PDF, reproducibility scripts, derived input packets, audit summaries, and the arXiv-style source archive.

The most important repository paths are:

- paper5_submission_source/: paper-facing draft, LaTeX facsimile, and PDF;
- figures/: rendered manuscript figures in PDF and PNG form;
- src/fmc/: scoring, covariance, coordinate, operator, and backreaction helper code;
- scripts/: executable reproduction and audit scripts;
- data/: derived public-input and local reproduction packets;
- evidence/: compact CSV outputs used by the manuscript and audit gates;
- docs/measurement_gate_audit_log.md: detailed implementation/audit log moved out of the main paper;
- frozen/: locked operator and configuration artifacts;
- wolfram/: independent Wolfram Language second-implementation checks;
- tests/: public reproducibility checks.

The key manuscript-facing outputs are:

- evidence/gate_results_current.csv;
- evidence/finite_memory_preflight_summary.csv;
- evidence/registered_protocol_guided_dominance_summary.csv;
- evidence/source_native_reproduction_family_dominance_summary.csv;

- studies/finite_memory_cosmology_paper5_v01/wolfram_audit_logs/finite_memory_diagnostic_check_v
- finite_memory_projection_corrections.pdf;
- arxiv_submission_source.zip.

The current comparison should be treated as reconstruction-level diagnostic preflight. Raw survey data and unpublished source-native derivative exports are not redistributed. Source-native derivative/covariance exports remain a blocker for measurement-validation language.

The current package can be regenerated with:

```
python3 -m pip install -r requirements.txt
PYTHONPATH=src python scripts/run_gate_current_packet.py
PYTHONPATH=src python scripts/build_finite_memory_preflight_packet.py
wolframscript -file wolfram/FiniteMemory_Diagnostic_Check.wl
python studies/finite_memory_cosmology_paper5_v01/make_paper5_submission_source_v01.py
python -m pytest -q
```

References

[1] A. Heinesen and T. Clifton, ”Observational Tests for Distinguishing Classes of Cosmological Models”, arXiv:2604.07244, 2026. https://arxiv.org/abs/2604.07244

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